

SEAT No. _____

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SARDAR PATEL UNIVERSITY
M.Sc.(IT) (SEMESTER - II) EXAMINATION
PS02CINT52: Software Engineering
Saturday, 15th April, 2023



Time: 10:00 a.m. to 01:00 p.m.

Max. Marks: 70

1.(A) Select the most appropriate option for each question :

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- (i) _____ refers to the difference between an actual output and an ideal output.
(A) A failure (B) An error (C) A resolution (D) None of these
- (ii) An SRS provides a reference for _____ of the final product.
(A) verification (B) validation (C) refinement (D) None of these
- (iii) UML stands for
(A) Unified Modeling Language (B) Universal Modeling Language
(C) Uniform Modeling Language (D) None of these
- (iv) An SRS establishes a basis for agreement between _____.
(A) a user and a product (B) a product and a quality
(C) a client and a supplier (D) developers and a project
- (v) In a use-case diagram, each use case is represented by _____.
(A) an ellipse (B) a circle (C) a rectangle (D) a diamond
- (vi) Which of the following diagrams can show a set of parallel or concurrent flows of activities?
(A) A class diagram (B) A use-case diagram
(C) A flow chart (D) An activity diagram
- (vii) _____ is typically done by the programmer of a module.
(A) Unit testing (B) Beta testing
(C) Black-box testing (D) None of these
- (viii) In a class diagram, each class is represented by _____.
(A) an oval (B) a circle (C) a rectangle (D) a diamond

2. Answer the following questions (ANY SEVEN):

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- (i) Define : *software engineering*. What are the *challenges* in software engineering?
- (ii) Differentiate between *functional* and *structural* testing.
- (iii) What is *agile computing*?
- (iv) What are the *advantages* of a *modular design*?
- (v) Which *criteria* can be used for determining the *quality* of *software design*?
- (vi) List *only the names* of different types of *UML diagrams*.
- (vii) Distinguish between module *coupling* and *cohesion*.
- (viii) Differentiate : *unit testing* vs *integration testing*.
- (ix) What is *regression testing*?

(1)

(P.T.O)

- 3.(A) Describe the *waterfall model* giving its limitations. 6
- 3.(B) Describe the *spiral model* giving its advantages. 6
- OR
- 3.(B) Describe the *prototyping model* giving its advantages and disadvantages. 6
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- 4.(A) Explain the classical *Systems Development Life Cycle (SDLC)* method for developing information systems. 6
- 4.(B) What is an SRS ? Describe the general characteristics of an SRS. 6
- OR
- 4.(B) Describe the major issues in *software project planning*. 6
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- 5.(A) Describe the features and advantages of *object-oriented design*. What do you mean by *information hiding* ? 6
- 5.(B) Explain the UML *use-case diagrams* giving an example. 6
- OR
- 5.(B) Explain the UML *class diagrams* giving an example. 6
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- 6.(A) Write a short note on *coding standards and coding guidelines*. 6
- 6.(B) Describe the characteristics of *white-box testing*. 6
- OR
- 6.(B) Write a short note on *test oracles*. 6

(2)